Questions

These questions use the following database table, WEATHER, that store weather readings.

Date	Time	Temperature	Wind speed	Humidity
11/12/2019	10:00	20.3	22	59
11/12/2019	13:00	23.7	15	63
11/12/2019	15:00	24.0	13	61
11/12/2019	18:00	23.9	16	60
12/12/2019	15:00	18.4	20	52
12/12/2019	18:00	17.2	18	48

20 Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
ORDER BY Temperature
```

21 Write the results returned by the following SQL script:

```
SELECT Date, Time, Temperature
FROM WEATHER
ORDER BY Humidity DESC
```

22 Complete the SQL script to display the date, wind speed and humidity in ascending order of wind speed.

```
..... Date, Wind speed, Humidity
..... WEATHER
..... Wind speed
```

23 Complete the SQL script to display the date, time and temperature, in descending order of humidity.

```
SELECT Date, Time, .....
FROM WEATHER
ORDER BY ..... .....
```

24 Complete the SQL script to display the temperature, wind speed and humidity in descending order of temperature.

```
..... ....., Wind speed, .....
FROM .....
..... ..... .....
```

9.6 SUM

SUM command will add up the field that is written after it. It is written in the select statement in the format:

```
SELECT SUM(field)
FROM table
```

You can also use `WHERE` conditions. e.g.

```
SELECT SUM(field)
FROM table
WHERE condition
```

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Example 1:

```
SELECT SUM(NumberItems)
FROM Orders
```

This query will return:

52

It is calculated from adding together all of the number of items (20 + 2 + 1 + 29).

Example 2:

```
SELECT SUM(TotalCost)
FROM Orders
WHERE Posted = No
```

This query will return:

81.64

It is calculated from adding together all the total costs where posted is No (21.99 + 59.65).

9.7 COUNT

This will count how many records meet the criteria, or how many fields are in the table.

Count is in the format:

```
SELECT COUNT(field)
FROM table
```

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Example 1:

```
SELECT COUNT(OrderID)
FROM table
```

The query will return:

4

Example 2:

```
SELECT COUNT(NumberItems)
FROM table
WHERE Posted = Yes
```

The query will return:

2

Example 3:

```
SELECT COUNT(OrderID)
FROM Orders
WHERE NumberItems > 10
```

The query will return:

2

SKILLS FOCUS 9.1

SUM AND COUNT

SUM and COUNT are both mathematical operations that perform different functions.

SUM

This performs addition of a group of values.

Example 1, a set of data.

1 + 3 + 8 + 9 + 3 = 24

Example 2, a table of data.

10
2
9
30
4
5

The SUM is 10 + 2 + 9 + 30 + 4 + 5 = 60

Example 3, an SQL query.

This query will use the following table, SALES.

CONTINUED

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

```
SELECT SUM(Total)
FROM SALES
```

The command word `SUM` tells you it needs you to add values together. The field in the brackets, `Total`, tells you to add together the Total column.

20.00 + 16.25 + 8.99 + 3.00 + 11.50 = 59.74

When you are answering questions such as what does this script return when `SUM` is being used, the only answer is the number. In Example 3, the answer to the question would just be: 59.74

Example 4, table with criteria using the table `SALES`.

```
SELECT SUM(Total)
FROM SALES
WHERE Customer ID = "2"
```

This time there is a criteria. Only records with Customer ID being 2 are needed.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

There are two records with Customer ID 2. The total for these two records are added together.

16.25 + 3.00 = 19.25

The SQL script will return:

19.25

COUNT

This counts the number of records (rows). This is not related to the values in the fields, just how many records there are that meet the criteria.

Example 1, a set of data.

1, 3, 8, 9, 3. There are **5** values.

CONTINUED

Example 2, a table of data.

10
2
9
30
4
5

There are **6** values

Example 3, an SQL query.

This query will use the following table, SALES.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	3	3.00
5	4	11.50

```
SELECT COUNT(Total)
FROM SALES
```

The command word COUNT tells you that you are counting records. The field in the brackets, Total, tells you which to count.

20.00, 16.25, 8.99, 3.00, 11.50

The value returned will be: 5

When you are answering questions such as what does this script return when COUNT is being used, the only answer is the number. In Example 3, the answer to the question would just be: 5

Example 4, table with criteria using the table SALES.

```
SELECT COUNT(Total)
FROM SALES
WHERE Customer ID = "2"
```

This time there is a criteria. Only records with Customer ID being 2 are needed.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

CONTINUED

Count how many records have Customer ID 2. There are 2 records.

The SQL script will return:

2

Questions

These will use the following table, PRODUCTS:

ID number	Name	Type	Cost	Quantity in stock
12GH	Saturn	Chocolate	0.56	100
59RF	Vanilla	Fudge	1.99	20
3WR	Peanuts	Nuts	2.56	2
52D	Jazz	Fruit	0.23	33
6SE	Signet	Fudge	1.48	5

1 What is the difference between SUM and COUNT?

2 What will be returned when the following SQL script is run?

```
SELECT SUM(Quantity in stock)
FROM PRODUCTS
```

3 What will be returned when the following SQL script is run?

```
SELECT COUNT(ID number)
FROM PRODUCTS
```

4 What will be returned when the following SQL script is run?

```
SELECT SUM(Cost)
FROM PRODUCTS
WHERE Quantity in stock > 10
```

5 What will be returned when the following SQL script is run?

```
SELECT COUNT(Name)
FROM PRODUCTS
WHERE Cost < 2.00
```

ACTIVITY 9.2

There are other mathematical functions that SQL can perform. Find at least one other example of a mathematical function and how to write an SQL script using it.

Take one of the example databases and write a script using your function.

Peer Assessment

Swap your new function and your script with a partner. Research the new function your partner has used and check whether their script is accurate. If not, discuss the error and how to solve it.

PROGRAMMING TASK 9.3

These questions will need your shop database table from Programming Task 9.1.

Getting started

1 Identify the commands to order a query in descending order.

2 Identify the commands to add together a set of values.

3 Identify the commands to count how many records there are.

Practice

1a Complete the SQL script to return the name of all the products in the store in descending order of quantity in stock.

```
SELECT .....
FROM .....
ORDER BY .....
```

1b Identify the values that will be returned from your query in **1a**.

2a Complete the SQL script to return the total cost of all products.

```
SELECT ..... (.....)
FROM .....
```

2b Identify the values that will be returned from your query in **2a**.

3a Complete the SQL script to return how many items have been ordered for more products.

```
SELECT ..... (.....)
FROM .....
WHERE ..... ..... .....
```

3b Identify the values that will be returned from your query in 3a.

4 Complete the SQL script to return how many items are red, and have more than 100 in stock, in ascending order by the number in stock.

```
SELECT ..... (.....)
FROM .....
WHERE ..... ..... .....
AND ..... ..... .....
..... .....
```

Challenge

This task goes beyond the range of the specification.

Write each of the queries in the practice section in your database and execute them to see the results.

REFLECTION

How did you approach Programming Task 9.3? Did you have other queries with you to check your syntax and to help you answer the questions? How will you make sure you remember the commands?

Questions

These questions use the following database table, ORDERS, that stores details about orders from a shop.

Order ID	FirstName	LastName	NumberItems	TotalCost
15	Keanu	Kawai	5	20.59
16	Keanu	Kawai	10	101.12
17	Eka	Loke	3	30.50
18	Keanu	Kawai	2	14.00
19	Ariana	Wang	4	21.10
20	James	Smith	6	44.60
21	James	Smith	20	156.90
22	Ajay	Gupta	1	10.00

25 Write the results returned by the following SQL script:

```
SELECT SUM(NumberItems)
FROM ORDERS
```

26 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
```

27 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
WHERE NumberItems > 10
```

28 Write the results returned by the following SQL script:

```
SELECT SUM(TotalCost)
FROM ORDERS
WHERE FirstName = "Keanu"
```

29 Complete the SQL script to display how many orders James Smith has made.

```
SELECT ..... (FirstName)

FROM ORDERS

WHERE FirstName = "James" ..... LastName =
"Smith"
```

30 Complete the SQL script to calculate the total cost of orders where 5 or more items have been ordered.

```
SELECT ..... (TotalCost)

FROM ORDERS

..... NumberItems ..... 5
```

31 Complete the SQL script to calculate how many orders cost more than 50.00

```
SELECT ..... (NumberItems)

FROM .....

WHERE ..... ..... .....
```